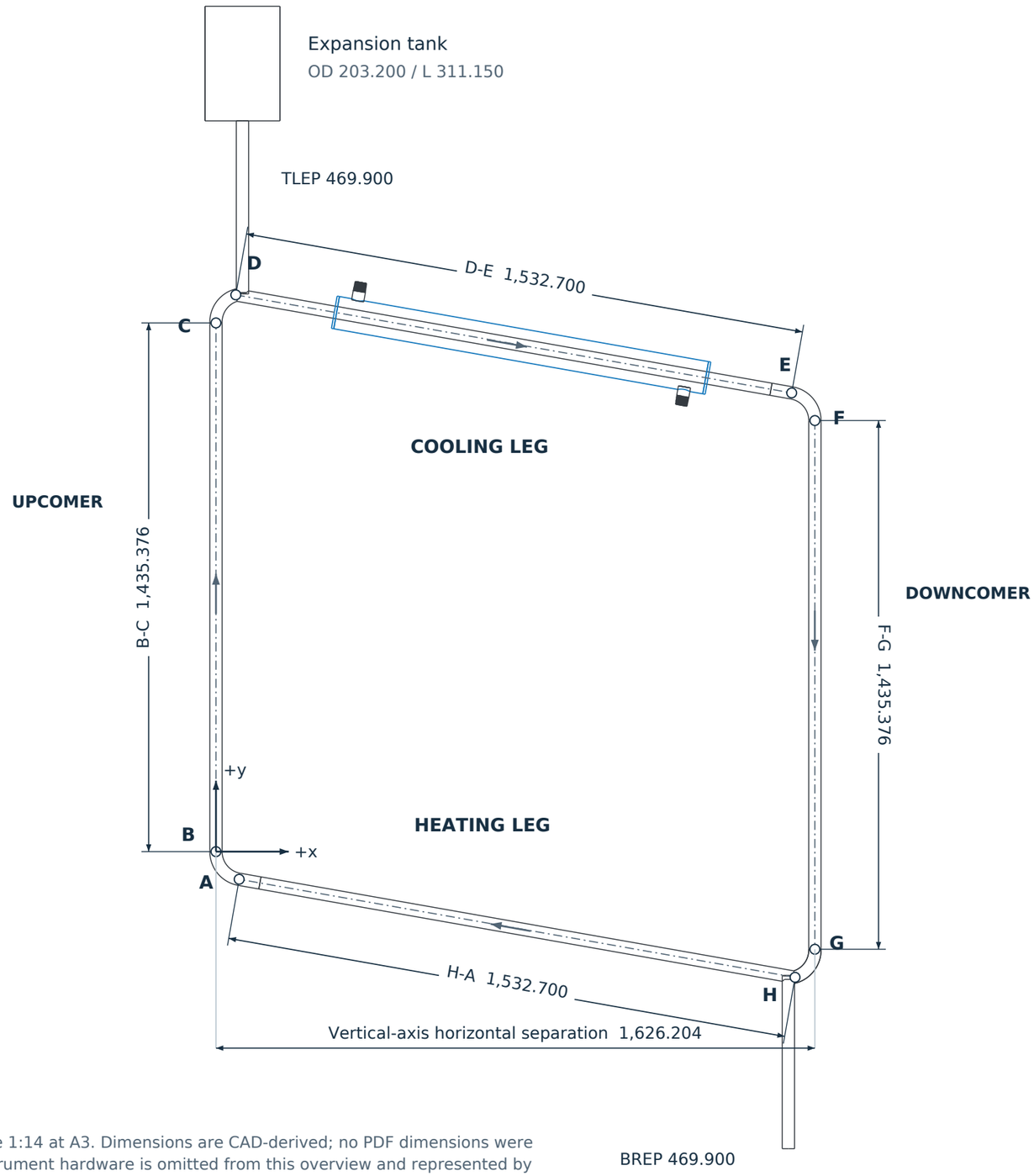
General arrangement and pipe spans

Front orthographic CAD projection. Tank at upper left. Clockwise traversal starts at B, the upcomer inlet.



View scale 1:14 at A3. Dimensions are CAD-derived; no PDF dimensions were used. Instrument hardware is omitted from this overview and represented by reference symbols on Sheet 5.

Measured straight spans

Leg / direction	Axis-end length	|dx|	|dy|
Upcomer B-C	1,435.376	0.000	1,435.376
Cooler D-E	1,532.700	1,509.415	266.151
Downcomer F-G	1,435.376	0.000	1,435.376
Heater H-A	1,532.700	1,509.415	266.151

Lengths above are between the extracted ends of the straight cylindrical axes. Curved corner lengths are listed separately on Sheet 2. Cooling and heating legs are inclined approximately 10 degrees to horizontal.

Datum and coordinate convention

B = (0, 0, 0) in the local drawing frame.
CAD B = (629.807, 852.124, 3,306.140) mm.
x = B_Z - CAD Z, rightward.
y = CAD Y - B_Y, upward.
n = CAD X - B_X, out of this drawing.

What is physically modeled

Blue outlines identify only the jacket shell and its two end plates. The bare cooling-leg pipe remains gray. The CAD does not separately define the active heater boundaries, so the heating-leg dimension is not a heated-length measurement.

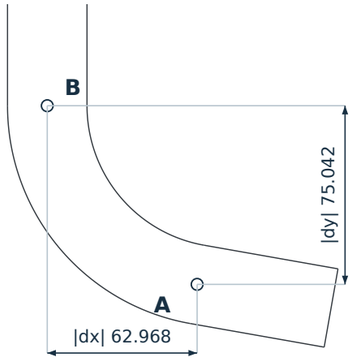
Drawing set

Sheet	Contents
2	Corner offsets, heights, virtual corner-to-corner spans
3	Cooler, jacket, ports, radii and diameters
4	Heating leg, exit tubes and tank interfaces
5	Sensor locations with nearest-end callouts
6	Sensor stations, distance tables and contact ambiguity
7	Ordered CAD and local coordinate schedule
8	CAD references, probe geometry and model-input checks

Corners, heights and end-to-end definitions

A-H identify two axial endpoints per physical corner. Virtual sharp intersections VLL/VUL/VUR/VLR are a separate modeling construction.

Lower left A-B

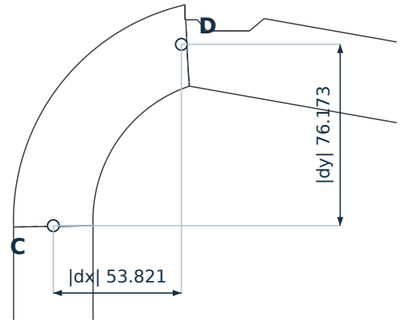


Chord 97.961

Arc 106.395 / R 76.200 / 80.000 deg

Envelope: inside R 59.4995 / outside R 92.9005

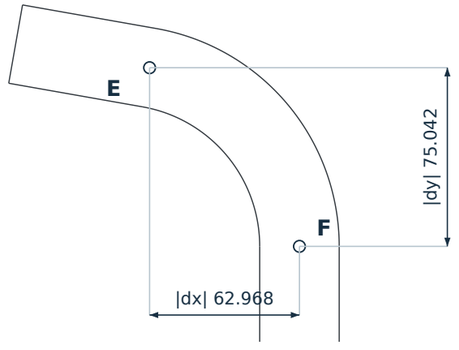
Upper left C-D



Chord 93.268

Recovered spine 99.578 (qualified below)

Upper right E-F

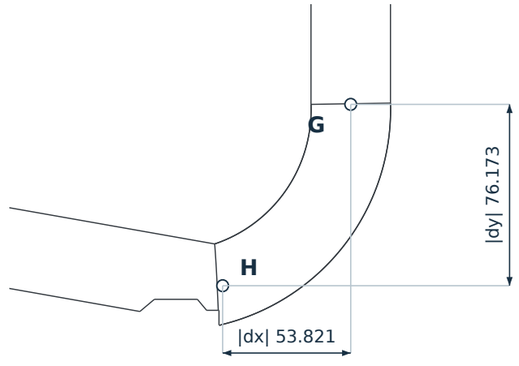


Chord 97.961

Arc 106.395 / R 76.200 / 80.000 deg

Envelope: inside R 59.4995 / outside R 92.9005

Lower right G-H



Chord 93.268

Recovered spine 99.578 (qualified below)

Signed segment changes in traversal order

Ends	Segment	dx	dy	dn	Length
B-C	Upcomer	0.000	1,435.376	0.000000	1,435.376
C-D	Upper Left Bend	53.821	76.173	0.000000	99.578*
D-E	Cooler	1,509.415	-266.151	0.271771	1,532.700
E-F	Upper Right Bend	62.968	-75.042	-0.010963	106.395
F-G	Downcomer	0.000	-1,435.376	-0.246119	1,435.376
G-H	Lower Right Bend	-53.821	-76.173	-0.014688	99.578*
H-A	Heater	-1,509.415	266.151	0.000000	1,532.700
A-B	Lower Left Bend	-62.968	75.042	0.000000	106.395

* C-D and G-H: 99.578 is a reconstructed spline-support spine, not an established exact joined hydraulic length. Its projected endpoints miss the cylinder limits by 0.009 and 1.031 mm. The exact endpoint chord is 93.268 mm. Do not silently substitute a circular elbow.

Endpoint heights relative to B

Point	x	Height y
B	0.000	0.000
C	0.000	1,435.376
D	53.821	1,511.549
E	1,563.236	1,245.398
F	1,626.204	1,170.356
G	1,626.204	-265.020
H	1,572.383	-341.193
A	62.968	-75.042

Highest listed tangent/axis limit: D +1,511.549.
Lowest: H -341.193.
Vertical range: 1,852.741 mm. This is a geometric height range, not an evaluated buoyancy head.

Virtual sharp-corner spans (projected axes extended to intersections)

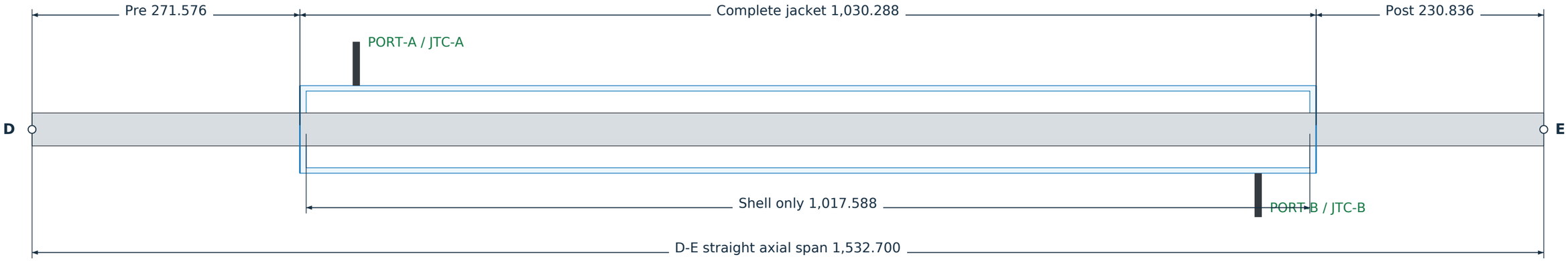
Leg	Virtual endpoints	dx	dy	Intersection-to-intersection length
Upcomer	VLL to VUL	0.000	1,584.978	1,584.978
Cooler	VUL to VUR	1,626.204	-286.744	1,651.291
Downcomer	VUR to VLR	0.000	-1,584.978	1,584.978
Heater	VLR to VLL	-1,626.204	286.744	1,651.291

SOURCE: UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

Units: mm | Drawing issue: 2026-09-26 | Printed precision does not imply manufacturing tolerance

Cooling leg, jacket and section dimensions

All axial dimensions run from D toward E. Complete jacket includes the two modeled end plates; effective thermal length is not specified.



Section diagram straightened to the cooling-leg axis. Gray core denotes modeled pipe envelope; its bore is absent from the source.

Radii and diameters from analytical CAD cylinders

Component	Outer radius	Inner radius	OD	ID / hole	CAD instance / face evidence
Main swept loop	16.7005	-	33.4010	-	NAUO1/NAUO45; NAUO2/NAUO45 / F3,5,6,9,11,12
TLEP / BREP	16.7005	13.4620	33.4010	26.9240	NAUO3; NAUO5 / F2 outer; 1 inner
Cooling jacket shell	44.4500	38.9636	88.9000	77.9272	NAUO6/NAUO40 / F4 outer; 3 inner
Jacket end plates	44.4500	16.8275	88.9000	33.6550	NAUO6/NAUO41; NAUO42 / F2 outer; 1 hole
Jacket port nipples	16.7005	13.3223	33.4010	26.6446	NAUO6/NAUO43; NAUO44 / F1 outer land; 27 bore
LF / combined LF-PTC spools	16.7005	13.3223	33.4010	26.6446	NAUO1/NAUO46/NAUO47; NAUO7/NAUO31 / F2/1; 5/4
PTC / JTC mounting spools	16.7640	12.7000	33.5280	25.4000	NAUO9,22,23,24/NAUO28 / F3 outer; 2 bore
TC sheath	1.5875	-	3.1750	-	NAUO7,8,9,22,23,24 / FSee shaft table / JSON
Expansion tank shell	101.6000	99.0469	203.2000	198.0938	NAUO4 / F2 outer; 1 inner

Missing radius/ID = not encoded, not zero. End-plate inner radius is the clearance-hole radius. Nipple outer radius refers to the cylindrical land. Local LF and PTC spool bores differ, so none is promoted to a loop-wide hydraulic diameter.

Jacket ends and coolant ports

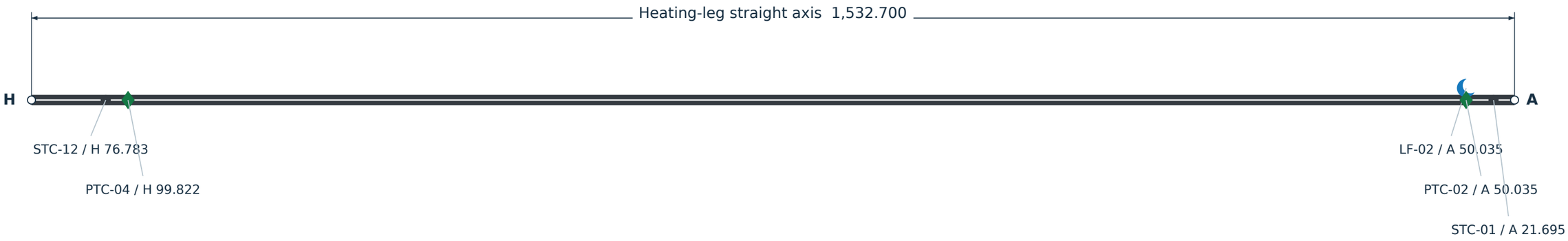
Feature	CAD X, Y, Z at axis	Station from D
Complete jacket upstream plane	629.807, 2,316.514, 2,984.869	271.576
Complete jacket downstream plane	629.807, 2,137.607, 1,970.234	1,301.863
PORT-A axis crossing of jacket axis	629.807, 2,306.590, 2,928.587	328.726
PORT-B axis crossing of jacket axis	629.807, 2,147.806, 2,028.079	1,243.126

Each end plate: 6.350 thick. Port-axis spacing: 914.400. Port A: 57.150 from jacket inlet plane. Port B: 58.738 from jacket outlet plane. Each port bore axis: 50.800 long.

Heating leg, exit tubes and tank interfaces

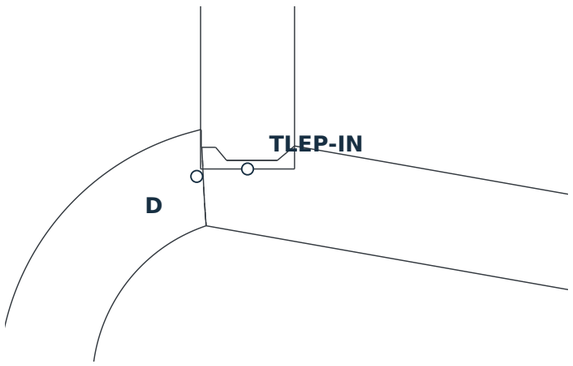
Exact endpoints refer to cylinder axes at the tube end planes. Branch connection points are not assumed to be ideal hydraulic tees.

Heating leg H to A



Straightened station diagram. Active heater start/end boundaries are not encoded in Rev8. This CAD extraction therefore does not assign a heater thermal length or color the whole heating leg as active heater.

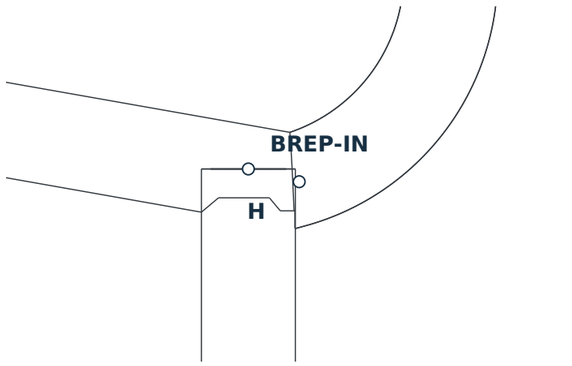
TLEP connection



IN relative to D: dx 18.072, dy 2.646

Out-of-plane dn 0.000000

BREP connection



IN relative to H: dx -18.072, dy 4.532

Out-of-plane dn 0.000231

Exit tube and tank axis lengths

Component	Axial length	OD	ID
TLEP	469.900	33.401	26.924
BREP	469.900	33.401	26.924
expansion_tank	311.150	203.200	198.094

TLEP-IN is the lower end nearest the loop; TLEP-OUT is nearest the tank. BREP-IN is the upper end nearest the loop; BREP-OUT is its lower end. These names describe the coordinate ordering, not an asserted branch flow direction.

TLEP-OUT lies 0.254 mm above the tank outer-cylinder lower end, a modeled axial overlap. BREP has 0.080572 mm change in CAD X along its 469.900 mm axis. The assembly has not been flattened or healed.

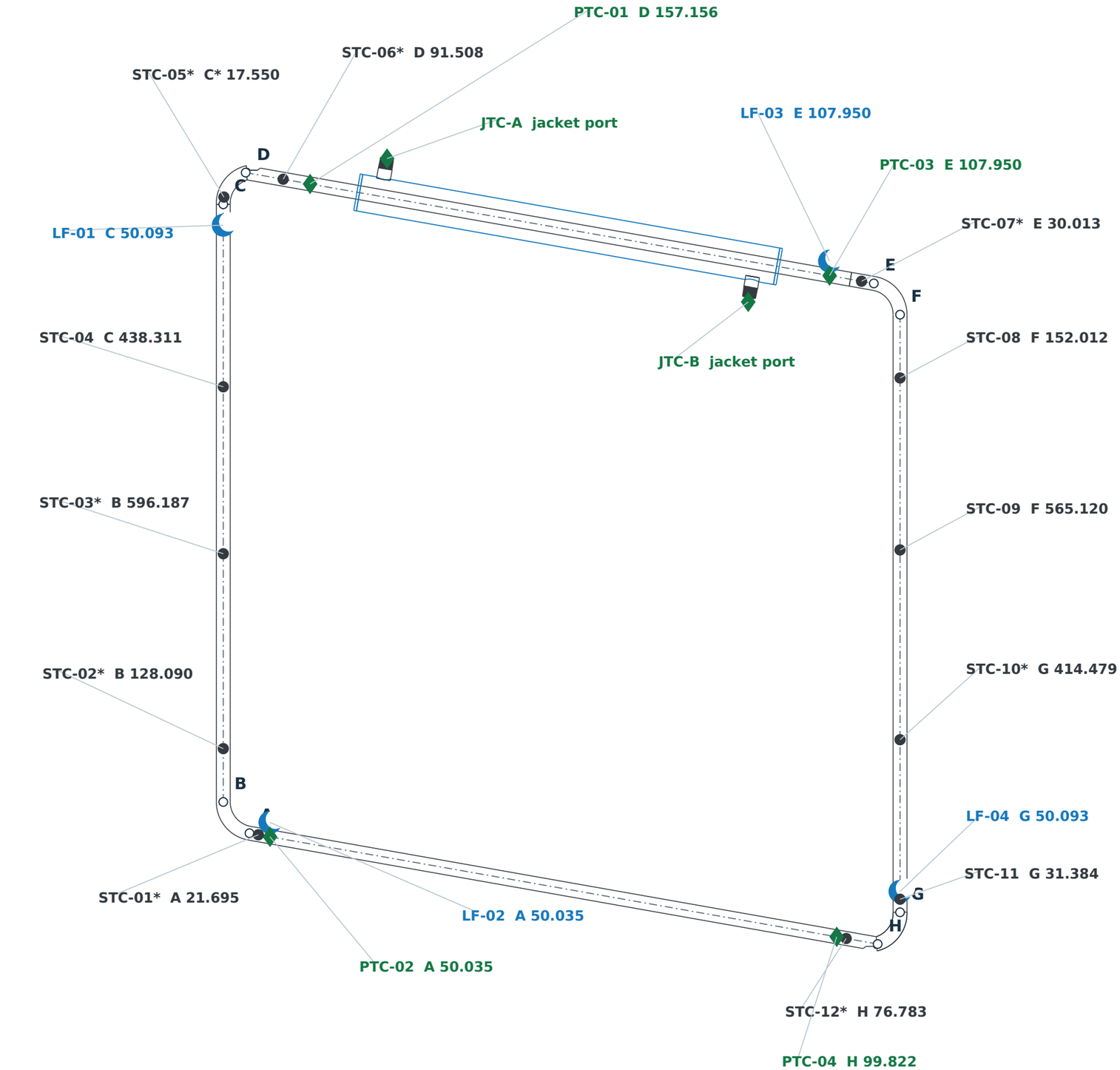
World-coordinate endpoint schedule

Axis endpoint	CAD X	CAD Y	CAD Z	CAD source
TLEP / IN	629.807	2,366.318	3,234.247	NAUO3 / F2
TLEP / OUT	629.807	2,836.218	3,234.247	NAUO3 / F2
BREP / IN	629.807	515.464	1,751.829	NAUO5 / F2
BREP / OUT	629.726	45.564	1,751.829	NAUO5 / F2
expansion_tank / IN	629.807	2,835.964	3,234.247	NAUO4 / F2
expansion_tank / OUT	629.807	3,147.114	3,234.247	NAUO4 / F2

For the tank, IN/OUT denote lower/upper shell-axis ends only. Its outside axial envelope is 311.150 mm. Source identity and face numbering are defined on Sheet 8.

Sensor locations and nearest-end distances

Symbols mark projected main-loop stations. Callouts give selected-reference distances along the leg; * identifies a non-unique or reconstructed reference.



Clockwise loop order: B-C-D-E-F-G-H-A-B. Nearest-end labels are along-axis distances, not horizontal or vertical separations. The jacket-only symbols use port references on Sheet 6.

Symbol and reference definitions

- LF - Luna fiber mount
- PTC - penetrating TC mount
- STC - surface contact reference

JTC-A/B use green diamonds and belong to the coolant ports. They are additional thermocouples, not missing main-loop PTC locations.

Main-loop dots and diamonds are placed on their projected pipe axes for clear station reading. The actual 3D mounting/contact coordinates are retained on Sheet 7. LF moons are slightly offset at shared LF/PTC mounts.

Reference integrity

STC markers use one closest point between the modeled thermocouple and pipe solid. Several bodies have multiple equally close points. Sheet 6 lists their station ranges; the CAD does not label a sensing junction.

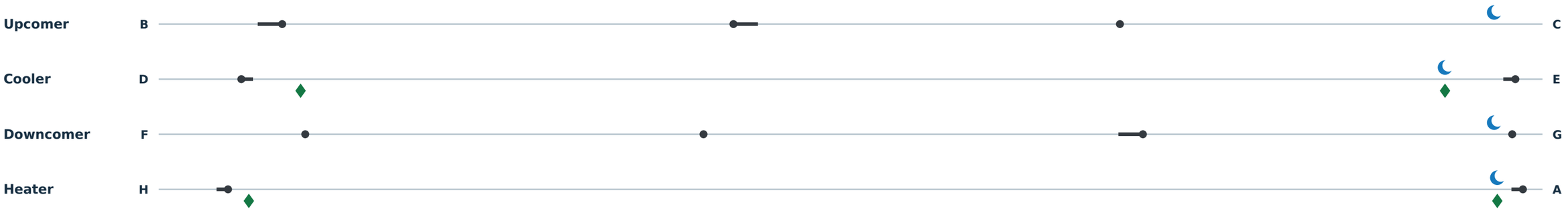
LF and PTC coordinates use the centroid of the local mounting-pipe body. These locate hardware, not the optical active length or TC junction.

Numbering and counts

12 STC + 4 LF + 4 loop PTC + 2 JTC.
LF-02/PTC-02 and LF-03/PTC-03 share mounting bodies. Labels follow the collaborator schema by region/order; STEP carries component instances, not verified acquisition-channel names.

Sensor stations and reference ambiguity

Distances use actual CAD references projected to the relevant leg. s is measured from the first endpoint in the leg name. No rounded-corner length is added.



Lane offsets separate sensor types in this unrolled view. Dark short bars span returned STC minimum-distance solutions; they are not certified sensing footprints.

Sensor	Leg ends	s from start	To end	Nearest end / distance	Other returned contact stations, measured from start
STC-02	B-C	128.090	1,307.286	B / 128.090	102.690 to 128.090
STC-03	B-C	596.187	839.190	B / 596.187	596.187 to 621.590
STC-04	B-C	997.065	438.311	C / 438.311	Single returned point
LF-01	B-C	1,385.283	50.093	C / 50.093	Mount centroid
STC-05	C*-D*	17.550	82.029	C* / 17.550	Recovered spline spine*
STC-06	D-E	91.508	1,441.192	D / 91.508	91.508 to 104.406
PTC-01	D-E	157.156	1,375.544	D / 157.156	Mount centroid
LF-03	D-E	1,424.750	107.950	E / 107.950	Mount centroid
PTC-03	D-E	1,424.750	107.950	E / 107.950	Mount centroid
STC-07	D-E	1,502.686	30.013	E / 30.013	1,489.277 to 1,503.199
STC-08	F-G	152.012	1,283.364	F / 152.012	Single returned point
STC-09	F-G	565.120	870.256	F / 565.120	Single returned point
STC-10	F-G	1,020.897	414.479	G / 414.479	995.497 to 1,020.897
LF-04	F-G	1,385.283	50.093	G / 50.093	Mount centroid
STC-11	F-G	1,403.992	31.384	G / 31.384	Single returned point
STC-12	H-A	76.783	1,455.916	H / 76.783	64.075 to 76.783
PTC-04	H-A	99.822	1,432.877	H / 99.822	Mount centroid
LF-02	H-A	1,482.665	50.035	A / 50.035	Mount centroid
PTC-02	H-A	1,482.665	50.035	A / 50.035	Mount centroid
STC-01	H-A	1,511.005	21.695	A / 21.695	1,498.301 to 1,511.005

* STC-05 is on the upper-left spline corner: recovered station 17.550 from C*, 82.029 to D*. Direct coordinate difference from C is dx +1.632 and dy +17.418 mm. C*/D* are projected spine endpoints with the residuals documented on Sheet 2.

Jacket TC	Port bore axis	From inner bore end	Beyond outer bore end	Off port axis	CAD assembly
JTC-A	PORT-A	50.840	0.040	0.066	NAUO9
JTC-B	PORT-B	61.742	10.942	0.066	NAUO22

SOURCE: UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

Units: mm | Drawing issue: 2026-09-26 | Printed precision does not imply manufacturing tolerance

Ordered coordinates for model construction

Traverse B to C to D to E to F to G to H to A, then return to B. Sensor rows contain actual CAD reference points, not their projected axis stations.

ID	Point basis	CAD X	CAD Y	CAD Z	Local x	Height y	Normal n	Segment / reference
B	Axis endpoint	629.807	852.124	3,306.140	0.000	0.000	0.000	upcomer
STC-02	Contact	646.491	980.214	3,305.412	0.728	128.090	16.685	upcomer
STC-03	Contact	645.778	1,448.311	3,311.022	-4.882	596.187	15.971	upcomer
STC-04	Contact	646.044	1,849.189	3,308.757	-2.617	997.065	16.237	upcomer
LF-01	Mount	629.807	2,237.407	3,306.140	0.000	1,385.283	0.000	upcomer
C	Axis endpoint	629.807	2,287.500	3,306.140	0.000	1,435.376	0.000	upper left bend
STC-05	Contact	646.470	2,304.918	3,304.508	1.632	1,452.794	16.663	upper left bend
D	Axis endpoint	629.807	2,363.673	3,252.319	53.821	1,511.549	0.000	cooler
STC-06	Contact	646.505	2,348.058	3,162.153	143.987	1,495.933	16.698	cooler
PTC-01	Mount	629.776	2,336.325	3,097.560	208.580	1,484.201	-0.031	cooler
LF-03	Mount	630.052	2,116.226	1,849.222	1,456.918	1,264.102	0.245	cooler
PTC-03	Mount	630.052	2,116.226	1,849.222	1,456.918	1,264.102	0.245	cooler
STC-07	Contact	646.778	2,102.534	1,772.497	1,533.643	1,250.410	16.971	cooler
E	Axis endpoint	630.078	2,097.522	1,742.904	1,563.236	1,245.398	0.272	upper right bend
F	Axis endpoint	630.067	2,022.480	1,679.936	1,626.204	1,170.356	0.261	downcomer
STC-08	Contact	646.711	1,870.465	1,678.920	1,627.220	1,018.340	16.904	downcomer
STC-09	Contact	645.704	1,457.358	1,675.065	1,631.075	605.233	15.897	downcomer
STC-10	Contact	646.552	1,001.580	1,681.101	1,625.039	149.456	16.746	downcomer
LF-04	Mount	629.830	637.197	1,679.936	1,626.204	-214.927	0.023	downcomer
STC-11	Contact	646.248	618.485	1,676.893	1,629.247	-233.639	16.441	downcomer
G	Axis endpoint	629.821	587.104	1,679.936	1,626.204	-265.020	0.015	lower right bend
H	Axis endpoint	629.807	510.931	1,733.757	1,572.383	-341.193	0.000	heater
STC-12	Contact	646.507	524.330	1,809.363	1,496.777	-327.794	16.700	heater
PTC-04	Mount	629.805	528.200	1,832.075	1,474.065	-323.924	-0.002	heater
LF-02	Mount	629.786	768.348	3,193.905	112.235	-83.776	-0.021	heater
PTC-02	Mount	629.786	768.348	3,193.905	112.235	-83.776	-0.021	heater
STC-01	Contact	646.507	773.296	3,221.810	84.330	-78.828	16.700	heater
A	Axis endpoint	629.807	777.082	3,243.172	62.968	-75.042	0.000	lower left bend
JTC-A	Mount	630.458	2,397.359	2,912.585	393.555	1,545.235	0.652	PORT-A
JTC-B	Mount	628.995	2,053.093	2,044.757	1,261.383	1,200.969	-0.812	PORT-B
J-IN	Jacket axis	629.807	2,316.514	2,984.869	321.271	1,464.390	0.000	Complete jacket outer end plane
J-OUT	Jacket axis	629.807	2,137.607	1,970.234	1,335.906	1,285.482	0.000	Complete jacket outer end plane
TLEP-IN	Tube axis	629.807	2,366.318	3,234.247	71.893	1,514.194	0.000	TLEP
TLEP-OUT	Tube axis	629.807	2,836.218	3,234.247	71.893	1,984.094	0.000	TLEP
BREP-IN	Tube axis	629.807	515.464	1,751.829	1,554.311	-336.661	0.000	BREP
BREP-OUT	Tube axis	629.726	45.564	1,751.829	1,554.311	-806.561	-0.080	BREP

Local x = 3306.139984198 - CAD Z; y = CAD Y - 852.124195141; n = CAD X - 629.806632391. Full-precision XYZ, projected sensor stations, additional contact solutions, probe-shaft endpoints and port-axis endpoints are included in the companion JSON/CSV package.

J-IN / J-OUT = complete jacket outer end planes at the axis. TLEP/BREP IN = tube end nearest the loop; OUT = remote end. Sensor coordinates represent geometric references as defined on Sheets 5-6, not verified sensing junctions.

CAD traceability and modeling qualifications

Source priority: Rev8 STEP geometry only. Earlier PDFs and Rev7 dimensions were not used to populate this drawing set.

Source identity and extraction

UTDTloopWithOpticsAndHXRev8(1).STEP
Assembly root: UTDTloopWithOpticsAndHXRev8
Source SHA-256:

```
c40a7e6ce597a35cdaa9d7358835b242
df408750b021385c1653d02a0871bf1e
```

Recursive XCAF placements were composed for all 88 assembly records. OCCT imported STEP inch units to millimetres. Native cylindrical/torus faces supplied axes, radii and bounds. CAD hidden-line projection supplied the drawing outlines.

An independent STEP reload checked 12 long main-pipe face bounds against projected boundary vertices: maximum endpoint discrepancy < 0.000001 mm. This checks extraction consistency, not manufacturing accuracy.

CAD instance references

Feature	Path below root
Upcomer / cooler / lower-left corner	NAUO1/NAUO45
Downcomer / heater / upper-right corner	NAUO2/NAUO45
TLEP / tank / BREP	NAUO3 / NAUO4 / NAUO5
Jacket shell / end plates	NAUO6/NAUO40 / NAUO41 / NAUO42
Jacket coolant ports	NAUO6/NAUO43 / NAUO44

Within each main-loop part: F3/F9 long inclined face; F5/F11 vertical face; F6/F12 short inclined continuation; F1/F13 exact torus; F4/F10 spline corner. Repeated faces are opposite halves, not extra pipe lengths.

Sensor identity to CAD assembly

Drawing IDs	Path below root
STC-01 / 02 / 03 / 04	NAUO10 / 13 / 12 / 11
STC-05 / 06 / 07 / 08	NAUO14 / 15 / 16 / 17
STC-09 / 10 / 11 / 12	NAUO18 / 19 / 20 / 21
LF-01 / LF-04	NAUO1/NAUO46 / NAUO2/NAUO46
LF-02 + PTC-02 / LF-03 + PTC-03	NAUO8 / NAUO7
PTC-01 / PTC-04	NAUO24 / NAUO23
JTC-A / JTC-B	NAUO9 / NAUO22

Mount-reference child: LF-01/04 -> NAUO47; combined LF/PTC -> NAUO31; PTC-only and JTC -> NAUO28. Surface references come from the whole STC part to the closest main-pipe solid. Source names and full paths are retained in JSON.

Model-input qualifications

Main bore / active heater

The solid main-loop envelope supplies OD 33.401 but no bore. Neither heater active length nor heat-flux boundaries are separately defined.

Spline corner closure

Recovered C-D and G-H spines are 99.578 long with 0.009 / 1.031 endpoint residuals. Exact hydraulic corner lengths remain unresolved.

Cooler assembly alignment

Two straight-axis pieces meet with a 0.271771 mm offset. The 1532.700 span sums their unique lengths; the offset is not added as pipe.

Sensor contact / junction

STC-08 and STC-11 minimum gaps are 0.005634 and 0.092102 mm. Multiple contact solutions are reported on Sheet 6. TC junctions and LF sensing extents are not identified.

Modeled thermocouple sheaths

Probe grouping	Longest shaft length	Shaft OD
PTC-01 / PTC-04	158.750	3.175
PTC-02 / PTC-03	152.400	3.175
JTC-A / JTC-B	158.750	3.175

These are geometric cylindrical-shaft spans. Both 3D shaft endpoints are supplied in JSON; neither endpoint is asserted to be the sensing junction.

Using the companion data

The JSON preserves full floating-point coordinates, evidence paths and qualified values. CSV tables provide ordered points, sensor stations, pipe segments and radii/diameters. Null/blank means unavailable. Retain the normal coordinate n for 3D modeling. Choose and document any bore, heater extent, sensor-junction or geometry-healing assumption before constructing the computational model.